

7 A number of diseases are associated with lifestyle risk factors.

Some of these risk factors cause mutations.

(a) Mutations can give rise to cancer.

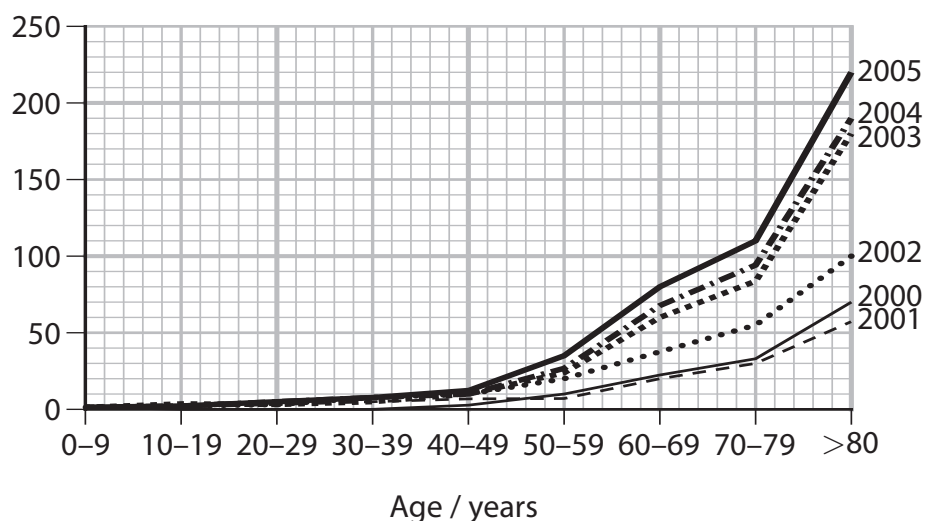
State the meaning of the term **mutation**.

(1)

(b) Exposure to ultraviolet light is associated with the development of skin cancer.

The graphs show the incidence of skin cancer in males and females in one country in the Far East, from 2000 to 2005.

Incidence of skin cancer per 100 000 of the male population



Incidence of skin cancer per 100 000 of the female population



- (i) The graphs show some correlations.

State the meaning of the term **correlation**.

(1)

- (ii) Describe the correlations shown by these graphs.

(3)

- (iii) Suggest a reason for each of the correlations shown by these graphs.

(3)



- *(c) The table shows some information from a study of the incidence of emphysema in smokers and non-smokers.

Information	Males	Females
Number of individuals in the study	25	25
Mean age when diagnosed / years	53.1	54.2
Range of ages when diagnosed / years	32 to 77	34 to 68
Number of smokers	5	6
Number of non-smokers	20	19
Number of smokers with emphysema	1	6
Number of non-smokers with emphysema	0	0

Criticise the design of this study.

(6)

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(Total for Question 7 = 14 marks)



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